



Incorporating Maths in Question-Writing

Matthew Rihan
Senior Project Mathematician

What is the Question for?

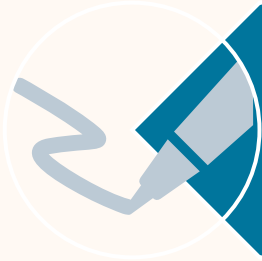


- What **purpose** is the question for?
 - Assessment
 - Practice to develop a skill
 - Extension to combine different skills
 - To highlight a particular case
- What **audience** is it for?
- These inform the **skills** required

Considerations for Maths Skills

- What **skills** are required?
 - Differentiation, chain rule, product rule, integration, partial fractions
- Are there multiple different **methods**?
- Do extra skills make an interesting **extension**?
 - Quadratics, indices, logarithms, binomial distribution
- Do extra skills add too much **challenge**?
 - Integration, double angle formulae, t-formulae
- How to **sequence** and develop the problem?
- Do we want to **model** an approach?
- **Work through** the problem to check.

Three Examples



Pure: Divisibility by Exhaustion



Mechanics: Pedal Power



Statistics: t-tests: Oak Leaves

Divisibility by Exhaustion



What **Maths skills** does it need?

https://isaacscience.org/questions/proof_divisibility_exhaustion

Divisibility by Exhaustion



Subject & topics

● Maths | Number | Arithmetic

Status

☐ Not started

Stage & difficulty

A Level Challenge 2 ☐ ☐ ☐

Further A Challenge 1 ☐ ☐ ☐

A sequence u_n is defined by $u_n = n^7 - n$, where $n \in \mathbb{N}$. The first four terms of this sequence are

0, 126, 2184, 16380, ...

What is the largest integer that will divide every term of this sequence?

Part A

Factorise u_n

Factorise u_n completely.

Divisibility by Exhaustion – A



Divisibility by Exhaustion



Subject & topics

● Maths | Number | Arithmetic

Status

☐ Not started

Stage & difficulty

A Level Challenge 2 ☒ ☐
Further A Challenge 1 ☒ ☐

A sequence u_n is defined by $u_n = n^7 - n$, where $n \in \mathbb{N}$. The first four terms of this sequence are

0, 126, 2184, 16380, ...

What is the largest integer that will divide every term of this sequence?

Part A

Factorise u_n ✓

Factorise u_n completely.

$$\begin{aligned} u_n &= n^7 - n \\ &= n(n^6 - 1) \\ &= n(n^3 - 1)(n^3 + 1) \\ &= n(n - 1)(n^2 + n + 1)(n + 1)(n^2 - n + 1) \end{aligned}$$

What **Maths skills** does it need?

- Factorising
- Difference of two squares
- Factor theorem
- Diff/sum of two cubes

Divisibility by Exhaustion – B, C, D



Part B

Divisibility by 2 ▾

Using your expression from part A, prove that every term in the sequence is divisible by 2.

Once you have worked out a proof yourself, consider the model proof below. Drag and drop answers into the blank spaces to complete the proof.

We know that the factors of u_n from part A.

When n is even, it is divisible by 2 and we can see that is a factor of u_n , so u_n is divisible by 2.

When n is odd, we can write $n = \text{$ in terms of k , where $k \in \mathbb{Z}$. Then the factor $\text{$ = $\text{$ in terms of k , so the factor $\text{$ is divisible by 2, and hence u_n is divisible by 2.

Therefore, u_n is divisible by 2 for any value of n . So every term in the sequence is divisible by 2.

Part C

Divisibility by 3 >

Part D

Divisibility by 7 >

What **Maths skills** does it need?

- Proof by exhaustion
- Split context into cases
- Represent numbers as a multiple, or off by an amount
- Substitution
- Divisibility

Divisibility by Exhaustion – E



Part E Largest Divisor ▾

Prove that 42 is the largest integer that will divide every term of u_n .

Once you have worked out a proof yourself, consider the model proof below. Drag and drop answers into the blank spaces to complete the proof. You may use the same answer more than once.

We know from earlier that u_n is divisible by 2, 3 and 7. So we know that $2 \times 3 \times 7 =$ will divide u_n . Are there any larger integers that can do so?

Let's consider the first non-zero term, 126. We find that $126 \div 42 =$. This shows that the prime factorisation of 126 is . Hence, the only larger factors of 126 are (in order of increasing size) and . Will these divide any other terms of u_n ?

Looking at the next term, we find that $2184 \div$ $= \frac{104}{3}$, so does not divide 2184. Considering our other factor, we find that $2184 \div$ $= \frac{52}{3}$, so does not divide 2184 either.

Therefore, 42 is the largest integer that will divide every term of u_n .

What **Maths skills** does it need?

- Proof by exhaustion
- Prime factorisation
- Divisibility

Divisibility by Exhaustion Summary



- Helps to **teach concept**
- Sequencing **establishes the steps** required
- Format helps to **model** writing a proof
- Multiple **methods** can be used

Pedal Power



What **Maths skills** does it need?

https://isaacscience.org/questions/pedal_power_r1

Pedal Power



Subject & topics

Physics | Mechanics | Dynamics

Status

☐ Not started

Stage & difficulty

A Level Challenge 2 ☒ ☒ ☐

A 75 kg cyclist on a 15 kg bicycle pedals against a backwards resistive force that is proportional to the square of their speed. On a flat road, they can travel at a steady speed of 10.0 m s^{-1} . While cycling up an incline, they produce the same power, but their steady speed is only 5.0 m s^{-1} .

Part A

Coasting down ▾

At what speed could the cyclist coast down the incline, if they do not pedal?

Pedal Power – A



Pedal Power



Subject & topics

Physics | Mechanics | Dynamics

Status

Not started

Stage & difficulty

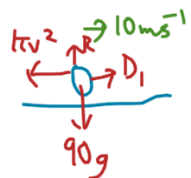
A Level Challenge 2

A 75 kg cyclist on a 15 kg bicycle pedals against a backwards resistive force that is proportional to the square of their speed. On a flat road, they can travel at a steady speed of 10.0 m s^{-1} . While cycling up an incline, they produce the same power, but their steady speed is only 5.0 m s^{-1} .

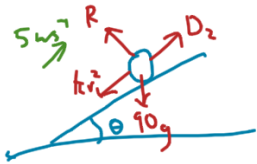
Part A

Coasting down ▾

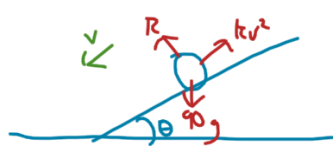
At what speed could the cyclist coast down the incline, if they do not pedal?



$$D_1 - kv^2 = 0$$
$$D = 1000 \text{ W}$$



$$D_2 - kv^2 - 90g \sin \theta = 0$$



$$kv^2 - 90g \sin \theta = 0$$
$$kv^2 = 90g \sin \theta$$

What **Maths skills** does it need?

- Modelling
- Draw diagrams
- Resolve forces
- Newton's laws
- Relate power and velocity
- Substitution

Pedal Power – B



Part B Head down ▾

The cyclist knows that, regardless of their speed, they can reduce the resistive force by 20% by putting their head down. This allows them to travel at a higher steady speed.

By how much does their speed increase, if they put their head down:

(i) while coasting down the incline?

 Unit ▾

(ii) while pedalling on a flat road?

 Unit ▾

What **Maths skills** does it need?

- Percentages
- Rearrange equations
- Substitution

$$\begin{aligned} k & \text{ becomes } 0.8k \\ 0.8kv^2 - 90g\sin\theta &= 0 \\ 0.8kv^2 &= 90g\sin\theta \\ 0.8kv^2 &= 175k \end{aligned}$$

Pedal Power – B (iii)



Part B

Head down ▾

The cyclist knows that, regardless of their speed, they can reduce the resistive force by 20% by putting their head down. This allows them to travel at a higher steady speed.

By how much does their speed increase, if they put their head down:

(i) while coasting down the incline?

 Unit ▾

(ii) while pedalling on a flat road?

 Unit ▾

(iii) while pedalling up the incline?

$$\begin{aligned}\frac{1000k}{v} - 0.8kv^2 - 90g\sin\theta &= 0 \\ 1000kv^{-1} - 0.8kv^2 - 175k &= 0 \\ 1000/k - 0.8v^3 - 175v &= 0 \\ -0.8v^3 - 175v + 1000 &= 0\end{aligned}$$

Part (iii) required solving a **cubic** in v with an irrational root. This was a skill we weren't intending to assess here.

Could use **numerical methods**, but we chose to **remove it**.

Pedal Power – C



Part C

Angle of the incline ▾

When the cyclist puts their head down while cycling on a flat road at a steady speed, their initial acceleration is 0.050 m s^{-2} .

What is the angle of the incline from earlier in the question?

$$\begin{aligned}v &= 10 \text{ m s}^{-1} \\ D_1 - 0.8kv^2 &= 90a \\ 100k - 0.8k(100) &= 90a \\ 100k - 80k &= 4.5 \\ 20k &= 4.5 \\ k &= \frac{9}{4}\end{aligned}$$

What **Maths skills** does it need?

- Newton's laws
- Rearrange equations
- Substitution
- Trigonometry

Pedal Power Summary



- Challenging question bringing together several skills
- Working through revealed a skill that was too much challenge
- Sequencing uses prior equations and asks what happens if this changes?

Example: t-tests: Oak Leaves



What **Maths skills** does it need?

https://isaacscience.org/questions/t_test_oak_leaves_r1

t-tests: Oak Leaves



Subject & topics

● Maths | Statistics | Hypothesis Tests

Status

☐ Not started

Stage & difficulty

Further A Practice 3 Three green hexagons representing a difficulty level of 3.

The leaves from oak trees growing in two different areas A and B are being measured. The lengths, in cm, of a random sample of 7 oak leaves from area A are:

6.2, 8.3, 7.8, 9.3, 10.2, 8.4, 7.2

Part A

Confidence interval ▾

Assuming that the distribution is normal, find a 95% confidence interval for the mean length of oak leaves from area A.

(Unit ▾ , Unit ▾)

Example: t-tests: Oak Leaves – A



t-tests: Oak Leaves



Subject & topics

● Maths | Statistics | Hypothesis Tests

Status

☐ Not started

Stage & difficulty

Further A Practice 3

The leaves from oak trees growing in two different areas A and B are being measured. The lengths, in cm, of a random sample of 7 oak leaves from area A are:

6.2, 8.3, 7.8, 9.3, 10.2, 8.4, 7.2

Part A

Confidence interval ▾

Assuming that the distribution is normal, find a 95% confidence interval for the mean length of oak leaves from area A.

(Unit ▾ , Unit ▾)

$$CI \quad \bar{x} \pm t \frac{s}{\sqrt{n}}$$

$$\bar{x} = \frac{\sum x}{7}$$
$$= 8.2$$

$$s_x^2 = \frac{1}{6} (\sum x^2 - 7 \bar{x}^2)$$
$$= 1 / (181.1 - 7 \times 67.24)$$

What **Maths skills** does it need?

- Mean
- Estimate population variance from sample
- Critical values for t-distribution
- Confidence intervals

Example: t-tests: Oak Leaves – B



Part B Assumptions ▾

The lengths, in cm, of a random sample of 5 oak leaves from area B are:

5.9, 7.4, 6.8, 8.2, 8.7

State two assumptions needed to carry out a suitable t -test on the difference between the mean lengths of oak leaves from areas A and B.

- ☐ The length of oak leaves from area B follow a geometric distribution.
- ☐ The lengths of oak leaves from areas A and B have equal means.
- ☐ The lengths of oak leaves from areas A and B have equal variances.
- ☐ The length of oak leaves from area B follow a normal distribution.
- ☐ The length of oak leaves from area B follow a uniform distribution.

What **Maths skills** does it need?

- Assumptions for two-sample t -test
- Requirements for t -distribution



$$\begin{aligned} s_p^2 &= \frac{1}{4} (\sum y^2 - 5 \bar{y}^2) \\ &= \frac{1}{4} (278.74 - 5 (7.4)^2) \\ &= \frac{147}{4} \\ &= 1.235 \end{aligned}$$

Summary Oak Leaves



- Standard question on the subject.
- Sequencing provides **information** as it becomes **relevant**, building from one-sample to two.
- Checks modelling **assumptions**.
- Format helps to **model** writing a hypothesis test.
- Multiple **methods** available, finding statistics manually or using a calculator.

Summary



- Establish the **skills** to assess and any related required skills.
- Consider whether there are multiple different **methods**.
- Consider whether extra skills make for interesting **extension** or **too much challenge**.
- Use **sequencing** to establish required steps and present information as it becomes relevant
- Can use to **model** an approach.
- **Work through** the problem to check.